

Rapid Parameter Inference for Spatiotemporal Stochastic Biological Models using Neural Posterior Estimation

Review

At the beginning of the review, I would like to mention that this is a topic of high scientific interest. Nevertheless, it is hard to see why it is not contributed to biological models. The basic problem is a question of modelling and simulation in general; the application in biology is just one of several fields of application. In the paper, you can understand that the method is in the focus, not the application to biological models. In that case, I would assume to pose a problem in the beginning, which will be solved in the end, after introducing the method. To compare, the scope of the journal, the proposed article could match to “Networks and Complex Systems”, but the scope tells furthermore: “Acceptable papers are those that bear significant importance on the biology per se being presented, and not on the mathematical analysis.” I would assume the presentation of a method alone is not satisfying this aspect of the scope.

The paper does not follow the traditional 'IMRaD' (Introduction, Methods, Results, and Discussion) format, which makes it difficult to relate if the paper is presenting the method or the application. Comparing the contribution to the biological model, it is minor to the part introducing the method. Furthermore, the distinction between “Discussion” and “Conclusion” is difficult to understand; it is usual that the conclusion is part of the discussion and would emphasise more the answer to the research question of the paper.

The upcoming numbering is introduced in the attached version of the paper. I structure my further review in order of this list:

1. Different abbreviations are multiply defined, e.g. see (1.). Furthermore, abbreviations are already defined in the abstract.
2. Some citations are strange, e.g. (2.). Is this intentional?
3. Figure 2: The sub-captions are unusual; the main caption should include the descriptions of the sub-captions (a)-(c).
4. Formulas should be integrated in the sentences and not introduced by a colon, when the sentence is continuing after.
5. Notation, framed in the text, is not defined.
6. Figure 5 is quite far from the first reference in the text.
7. In this paragraph, some biological problems, where the method would be applied, are mentioned. I would expect that these problems are the research question and the presented method is used to solve this. In the available article it looks vice versa.
8. Again, the biological models are “just” a side comment.

To summarise, the contribution is a good one, but I doubt if the journal is well selected. In the attached version of the paper, minor typos are highlighted and suggested to the authors.